ANX-PR/CL/001-01
LEARNING GUIDE

SUBJECT

83000024 - AmpliaciÓn De MatemÁticas

DEGREE PROGRAMME

08IN - Master Universitario En Ingenieria Naval Y Oceanica

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

Index

Learning guide

1. Description

1.1. Subject details

Name of the subject	83000024 - Ampliación de Matemáticas
No of credits	5 ECTS
Type	Compulsory
Academic year ot the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	08IN - Master Universitario en Ingenieria Naval y Oceanica
Centre	08 - E.T.S. De Ingenieros Navales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Fabricio Macia Lang (Subject coordinator)	P1-04	fabricio.macia@upm.es	M - 12:30 - 14:00 M - 15:30 - 19:00
Daniel Sanchez Mendoza		daniel.sanchezmen@upm.es	Sin horario.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Ordinary Differential Equations (Undergraduate level)

- Linear Algebra (Undergraduate level)

- Basic programming (Undergraduate level)

- One-variable and multivariate Calculus

4. Skills and learning outcomes *

4.1. Skills to be learned

CG1 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio.Students should be able to apply their acquired knowledge and problem-solving skills in new or unfamiliar environments within broader (or multidisciplinary) contexts related to their area of study.

CG2 - Que los estudiantes sean capaces de integrar conocimientos y enfrentarse a la complejidad de formular juicios a partir de una información que, siendo incompleta o limitada, incluya reflexiones sobre las responsabilidades sociales y éticas vinculadas a la aplicación de sus conocimientos y juicios. That students are able to integrate knowledge and face the complexity of making judgments based on incomplete or limited information, including reflections on the social and ethical responsibilities linked to the application of their knowledge and judgments.

CG3 - Que los estudiantes sepan comunicar sus conclusiones- y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades.That students know how to communicate their conclusions - and the ultimate knowledge and reasons that support them - to specialized and non-specialized audiences in a clear and unambiguous manner.

CG4 - (S1) Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CTUPM01 - (S2) Creatividad. Los estudiantes deben resolver de forma nueva, original y aportando valor, situaciones o problemas en el ámbito de la ingeniería.Creativity. Students must solve situations or problems in the field of engineering in a new, original and valuable way.

CTUPM04 - (S5) Uso de la lengua inglesa. Los estudiantes establecen conversaciones con nativos sin tener problemas de comunicación adicionales tanto de forma oral como escrita.Use of the English language. Students establish conversations with native speakers without having additional communication problems both orally and in writing.

CTUPM05 - (S6) Uso de las tecnologías de la información y comunicación (TIC). Los estudiantes aplican conocimientos tecnológicos necesarios de manera que les permitan desenvolverse cómodamente y afrontar los retos que la sociedad les va a imponer en su quehacer profesional empleando la informática.Use of information and communication technologies (ICT). Students apply the necessary technological knowledge in a way that allows them to perform comfortably and meet the challenges that society will impose on them in their professional work using information technology.

CTUPM06 - (S7) Comunicación oral y escrita. Los estudiantes transmiten conocimientos y expresan ideas y argumentos de manera clara, rigurosa y convincente, tanto de forma oral como escrita, utilizando los recursos gráficos y los medios necesarios adecuadamente y adaptándose a las características de la situación y de la audiencia.Oral and written communication. Students transmit knowledge and express ideas and arguments in a clear, rigorous and convincing manner, both orally and in writing, using the necessary graphic resources and media appropriately and adapting to the characteristics of the situation and the audience.

CTUPM08 - Trabajo en equipo. Los estudiantes desarrollan la capacidad para trabajar en equipo, integrarse y colaborar de forma activa en la consecución de objetivos comunes.Teamwork. Students develop the ability to work in teams, integrate and collaborate actively in the achievement of common objectives.

CTUPM09 - Resolución de problemas. Los estudiantes son capaces de identificar o proponer un problema, y tienen el conocimiento sobre diferentes alternativas metodológicas y estratégicas para resolverlo.Problem solving. Students are able to identify or propose a problem, and have the knowledge about different methodological and strategic alternatives to solve it.

CTUPM13 - Trabajo en contextos internacionales. Los estudiantes son capaces de integrarse en un grupo o equipo, colaborando y cooperando con otros. Tienen la capacidad para trabajar con estudiantes de otras disciplinas y de aceptar la diversidad social y cultural.Work in international contexts. Students are able to integrate into a group or team, collaborating and cooperating with others. They have the ability to work with students from other disciplines and to accept social and cultural diversity.

4.2. Learning outcomes

RA31 - C: Capacidad para resolver problemas complejos y para tomar decisiones con responsabilidad sobre la base de los conocimientos científicos y tecnológicos adquiridos en materias básicas y tecnológicas aplicables en la ingeniería naval y oceánica, y en métodos de gestión

RA32 - C: Capacidad para la resolución de los problemas matemáticos que puedan plantearse en la ingeniería y aptitud para aplicar los conocimientos adquiridos

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

This course is an introduction to Scientific Computing, with particular emphasis in Engineering applications.

5.2. Syllabus

1. The Python programming language and Jupyer notebooks

2. Error Analysis

3. Interpolation of data and functions

4. Polynomial approximation

5. Numerical quadrature

6. Numerical differentiation

7. Numerical solution of non-linear equations: root finding and optimization

8. Numerical integration: solving initial value problems for Ordinary Differential Equations

9. Métodos numéricos de resolución de problemas de conNumerical methods for solving boudary value problems for Ordinary Differential Equations

10. Introduction to the Finite Element Method

11. Introduction to Spectral / Fourier Analysis

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	**Classroom: theory and practice** Duration: 04:00 Additional activities			
2	**Classroom: theory and practice** Duration: 04:00 Additional activities			
3	**Classroom: theory and practice** Duration: 04:00 Additional activities			
4	**Classroom: theory and practice** Duration: 04:00 Additional activities			
5	**Classroom: theory and practice** Duration: 04:00 Additional activities			
6	**Classroom: theory and practice** Duration: 02:00 Additional activities			
7	**Classroom: theory and practice** Duration: 02:00 Additional activities **Test 1** Duration: 02:00 Additional activities			**Test 1** Problem-solving test Progressive assessment Presential Duration: 02:00
8	**Classroom: theory and practice** Duration: 04:00 Additional activities			
9	**Classroom: theory and practice** Duration: 04:00 Additional activities			
10	**Clase teórico-práctica** Duration: 04:00 Additional activities			
11	**Classroom: theory and practice** Duration: 04:00 Additional activities			
12	**Classroom: theory and practice** Duration: 04:00 Additional activities			

13	**Classroom: theory and practice** Duration: 04:00 Additional activities			
14	**Test 2** Duration: 02:00 Additional activities			**Test 2** Problem-solving test Progressive assessment Presential Duration: 02:00
15				
16				
17				**Final exam** Problem-solving test Global examination Not Presential Duration: 03:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
7	Test 1	Problem-solving test	Face-to-face	02:00	40%	/ 10	CG3 CG4 CTUPM01 CTUPM04 CTUPM05 CTUPM06 CTUPM08 CTUPM09 CG1 CG2 CTUPM13
14	Test 2	Problem-solving test	Face-to-face	02:00	60%	3 / 10	CG1 CG2 CG3 CG4 CTUPM01 CTUPM04 CTUPM05 CTUPM06 CTUPM08 CTUPM09 CTUPM13

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Final exam	Problem-solving test	No Presential	03:00	100%	5 / 10	CG1 CG2 CG3 CG4 CTUPM01 CTUPM04 CTUPM05 CTUPM06 CTUPM08 CTUPM09 CTUPM13

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Make up exam/Resit	Problem-solving test	Face-to-face	03:00	100%	5 / 10	CTUPM01 CTUPM04 CG1 CG3 CTUPM08 CTUPM09 CTUPM13 CTUPM05 CG2 CG4 CTUPM06

7.2. Assessment criteria

In order to obtain a pass grade, students must obtain at least 5 over 10 in any of the two following options:

Option 1. 40% Test 1 + 60% Test 2. In order to obtain pass grade following this path, students must obtain at least 3/10 in Test 2.

Option 2. 100% Final exam.

All tests and exams will consist in answering a number of questions in a Jupyter notebook. Some questions will be of a more theoretical/conceptual nature (the answer should be given in a text cell using the Markdown language); some will involve producing/assessing code (the answer should be given in a code cell and produce a fully functional code). Notes, books and any other class material that can be carried in an USB drive is allowed in exams; but access to the internet will be restricted.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
P. Angulo, F. Macià and D. Sánchez-Mendoza. Scientific Computing for Engineers (2025).	Bibliography	A course based on Jupyter notebooks that is freely available at https://framagit.org/pang/scicompeng
R.L. Burden, J.D. Faires, Análisis Numérico, 9ª edición. Cengage Learning, México (2011)	Bibliography	
D. Eriksson, K. Estep, C. Johnson, Applied Mathematics: body and soul. Volúmenes 1-5. Springer (2004)	Bibliography	
G. Farin, Curves and Surfaces for CAGD: a Practical Guide. 5ª edición. Morgan Kaufmann Publishers, San Francisco (2002)	Bibliography	
P. Henrici, Elementos de Análisis Numérico, Trillas, México (1972)	Bibliography	
D. Kincaid, W. Cheney, Análisis numérico : las matemáticas del cálculo científico, Addison Wesley, Buenos Aires (1994)	Bibliography	
J.M. Sánchez, A. Souto, Problemas de Cálculo Numérico para ingenieros con aplicaciones Matlab, Schaum McGraw-Hill, Madrid (2005)	Bibliography	
J.M. Sanz Serna, Diez lecciones de cálculo numérico, 2ª edición, Universidad de Valladolid, Valladolid (2010)	Bibliography	

J. Stoer, R. Bulirsch, R. Bartels, W. Gautschi, Introduction to numerical analysis, 2ª edición, Springer-Verlag, New York (1996)	Bibliography	
C.Vázquez, Cálculo numérico, Gª Maroto Editores, Madrid (2012)	Bibliography	
Aulas/Centro de Cálculo/Biblioteca/Salas de estudio	Equipment	
http://moodle.upm.es	Web resource	
https://github.com/mandli/intro-numerical-methods	Web resource	A course on Numerical Methods in Python, (developed for the Columbia course APMA 4300)